MTH 200 **Chapter 4: Applications of Derivatives**

4.4 The Mean Value Theorem Worksheet

1. Let $f(x)=2x^2+7$. Determine the value of c which satisfies the conclusion of the *Mean Value Theorem* on the interval $[1,5]$. You must show all work and simplify your answer.

2. Prove, using the *Intermediate Value Theorem* and *Rolle's Theorem*, that the function $f(x)=\frac{1}{3}x^3+2x^2+5x-1$ has exactly one real root.

3. Verify that $f(x) = e^{-2x}$, $[0,3]$ satisfies the hypothesis of the Mean Value Theorem. Then find all c that satisfy the conclusion of the Mean Value Theorem.

4. Verify that the hypothesis of Rolle's Theorem are satisfied for the function $f(x) = \sqrt{x} - \frac{1}{3}x$ on the interval $[0,9]$, and find all values of c in that interval that satisfy the conclusion of the theorem.

5. Let $f(x) = \tan(x)$. Show that $f(0) = f(\pi)$ but there is no number c in the interval $(0, \pi)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem?

6. Use Rolle's Theorem to prove that the function $f(x) = x + \frac{1}{2}\sin(x) - 3$ has exactly one real root.

7. Use Rolle's Theorem to prove that the function $f(x) = x^4 + 4x + 1$ has at most two real roots.

8. You are driving on a straight highway with a speed limit is 55 *mi/h*. At 8:05am a police car clocks your velocity at 50 *mi/h* and at 8:10am a second police car posted 5 miles down the road clocks your velocity at 55 *mi/h*. Explain why the police have the right to charge you with a speeding violation?